

GAYLORD 9 SSW, MI, USA

WMO: 725410

Lat: 44.9080N

Lon: 84.7203W

Elev: 445

StdP: 96.09

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 54854

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.0	(c) -17.5	(d) -24.2	(e) 0.4	(f) -19.8	(g) -21.3	(h) 0.6	(i) -17.4	(j) 5.7	(k) -4.2	(l) 5.2	(m) -5.1	(n) 1.6	(o) N/A	(p) 0.491

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.6	(c) 29.1	(d) 20.2	(e) 27.6	(f) 19.3	(g) 26.1	(h) 18.4	(i) 21.9	(j) 26.8	(k) 21.0	(l) 25.5	(m) 20.1	(n) 24.1	(o) 2.3	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 20.5	(b) 16.0	(c) 24.5	(d) 19.6	(e) 15.1	(f) 23.2	(g) 18.7	(h) 14.3	(i) 22.2	(j) 66.5	(k) 26.8	(l) 62.9	(m) 25.5	(n) 59.5	(o) 24.0	(p) 24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.9	4.3	3.8	-24.1	31.6	3.5	1.2	-26.6	32.5	-28.6	33.2	-30.5	33.9	-33.0	34.7
			-25.0	23.6	3.7	0.8	-27.6	24.2	-29.8	24.7	-31.9	25.1	-34.6	25.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-7.9	-7.2	-2.0	4.8	12.4	16.6	19.2	18.5	14.9	8.1	1.2	-4.5
	DBStd	10.94	5.80	5.81	6.60	5.62	5.32	3.79	3.20	3.21	4.38	5.03	5.44	4.95
	HDD10.0	2454	555	483	379	175	36	2	0	0	7	99	269	449
	HDD18.3	4567	813	716	632	407	197	76	29	37	121	318	514	707
	CDD10.0	1084	0	0	6	18	109	201	284	264	153	42	5	0
	CDD18.3	155	0	0	1	1	12	24	55	43	17	2	0	0
	CDH23.3	1150	0	0	5	6	126	186	414	294	104	14	0	0
	CDH26.7	225	0	0	1	0	21	37	85	59	22	1	0	0
Wind	WSAvg	1.9	2.1	2.2	2.3	2.4	2.1	1.6	1.4	1.3	1.4	1.8	2.0	2.1
Precipitation	PrecAvg	837	56	41	43	69	81	90	82	81	83	87	71	53
	PrecMax	995	92	68	115	102	135	179	196	140	140	220	137	113
	PrecMin	705	18	13	10	21	20	29	33	42	23	29	29	17
	PrecStd	89	22	15	24	24	29	38	36	29	29	44	33	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	9.9	22.9	24.1	28.9	29.9	31.0	30.4	29.4	25.5	18.4	11.4
		MCWB	7.1	7.5	15.6	12.9	19.1	21.3	21.4	20.2	20.7	15.4	12.2	9.8
	2%	DB	4.8	5.6	14.9	20.4	26.8	27.6	29.0	28.5	26.3	21.7	14.4	6.3
		MCWB	3.4	3.3	11.2	12.3	16.7	18.9	20.6	19.3	19.1	15.9	10.3	4.9
	5%	DB	2.0	2.7	10.2	16.8	24.4	25.4	27.3	26.5	23.9	18.8	11.4	3.9
		MCWB	0.6	1.1	6.5	9.9	15.8	18.0	19.8	18.7	18.0	15.0	8.3	2.7
	10%	DB	0.2	0.3	6.8	13.9	21.8	23.7	25.7	24.8	21.9	15.9	8.8	1.9
		MCWB	-0.6	-1.1	3.6	8.0	14.4	16.6	19.0	17.9	17.6	12.9	6.5	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	8.3	15.5	15.7	20.4	22.8	23.8	21.8	22.4	18.9	13.3	11.1
		MCDB	7.9	8.9	22.1	19.2	26.3	28.0	29.1	26.4	27.8	20.7	15.9	12.0
	2%	WB	3.1	3.9	12.4	12.6	18.9	20.6	22.5	20.9	20.8	17.0	10.5	5.7
		MCDB	4.7	5.6	15.6	16.7	24.0	25.5	27.1	26.0	24.2	19.3	13.6	6.3
	5%	WB	0.7	1.5	6.9	10.0	17.5	19.3	21.4	20.1	19.7	14.9	8.5	3.3
		MCDB	1.6	3.1	9.8	14.1	22.4	23.1	25.7	24.4	22.4	18.0	11.0	4.2
	10%	WB	-0.4	-0.5	3.7	7.6	15.8	18.2	20.3	19.2	18.5	12.8	6.4	1.5
		MCDB	0.4	0.5	6.3	11.9	20.2	22.1	24.4	22.9	21.6	15.3	8.4	2.4
Mean Daily Temperature Range	5% DB	MDBR	6.2	7.3	8.7	9.8	11.5	10.7	10.6	10.3	10.2	8.3	6.6	5.2
		MCDBR	7.6	9.3	11.4	14.5	13.9	12.3	12.2	12.3	11.7	12.3	10.5	8.0
		MCWBR	6.2	7.4	7.3	8.7	7.1	6.5	5.8	5.3	5.6	7.3	7.2	6.7
	5% WB	MCDBR	6.9	9.0	10.2	12.4	11.9	10.1	10.2	9.9	9.9	11.3	10.1	7.7
		MCWBR	6.0	7.8	7.3	8.8	6.7	6.2	5.8	5.2	5.7	8.0	7.7	7.1
Clear-Sky Solar Irradiance	taub	0.277	0.285	0.305	0.336	0.364	0.382	0.380	0.371	0.358	0.324	0.293	0.267	
	taud	2.325	2.290	2.413	2.385	2.366	2.333	2.357	2.376	2.412	2.507	2.527	2.430	
	Ebn at Noon	838	901	933	925	905	885	883	879	862	851	828	824	
	Edh at Noon	78	98	100	113	119	123	119	113	100	79	65	64	
All-Sky Solar Radiation	RadAvg	1.23	2.15	3.56	4.56	5.51	6.03	6.09	5.17	4.04	2.39	1.33	0.93	
	RadStd	0.12	0.23	0.40	0.44	0.49	0.41	0.27	0.31	0.36	0.28	0.19	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page